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Brief Synopsis of a Few Selected Projects

Deep Learning for Self-Driving Car

Nov 2016 – Present (expected completion: Feb 2017), Self-Driving Car Nanodegree, Coursera
Keywords: Deep Learning, Behavioral Cloning, Transfer Learning, Computer Vision
Technology: Python, TensorFlow, Keras, Pandas
Number of Team Members: 1
Description: The project is divided in various sub-tasks including detection of traffic lanes, identification of moving vehicles, classification of traffic signs using *Deep Learning* and *Computer Vision* techniques. The project also involves use of *Deep Neural Network* for *Behavioral Cloning* to imitate driving behavior. Currently, I am working on lane-detection and traffic-sign identification.

Predicting Building Abandonment

Nov 2016 – Present (expected completion: Dec 2016), Data Science Specialization, Coursera
Keywords: Machine Learning,
Technology: Python, scikit-learn, pandas
Number of Team Members: 1
Description: The project involves predicting building abandonment ("blight") based on several public datasets including criminal incidents and permit violation. The datasets include latitude and longitude, which have been grouped together for identification of building. The grouping was accomplished by using *Google Reverse Geo-Coding API*, however considering millions of records to be fetched, the building had been approximated as 6-char *geohash* corresponding to the given latitude-longitude pair of incidents. The labelled dataset for training had been generated using blight-incidents dataset having incident marked as 'Dismantle'. Currently, I am working on feature engineering for preparing training dataset.

Predict (Predicting Users for Advertising)

Nov 2016 – Present, Media IQ Digital India Ltd., Bengaluru
Keywords: Machine Learning, Naïve Bayes, Random Forest, K-Means Clustering, Behavioral Targeting, Big Data
Technology: Hive, Spark MLlib, Spark Streaming, Kafka
Number of Team Members: 4
Description: The project aims at predicting users who would take desirable action after an advertisement is displayed to them, and target them in real time. The current model is powered by *Naives Bayes Classifier* built by aggregating user's browsing behavior and geo-location. We are under the process of migrating the model to *Random Forest* which have provided about 10% uplift in preliminary analysis. To combat memory issues stemming out of the large feature set while using *Random Forest*, we grouped the features by *K-Means Clustering*, and used aggregated count of the group to train the model. Moving forward we would be incorporating temporal behavior in the model's feature set, and move to real-time prediction using Spark Streaming.

Predictive Client-side Profiling for Behavioral Advertising

Jun 2016 – Present, Media IQ Digital India Ltd., Bengaluru
Keywords: Real-time Prediction, Machine Learning, Client-side Profiling, Behavioral Targeting, Ad-exchange Architecture
Technology: JavaScript, HTML, Python, Kafka
Number of Team Members: 3
Description: The idea was harnessed as part of a Hackathon at Media IQ Digital India Ltd. Leveraging know-how of Web Development and Data Science, I proposed a novel scalable approach for client-side user profiling to offer personalized advertising experience. The proposed methodology's merit includes near-real-time user profiling, high scalability, low-bandwidth requirement, privacy issues redressal, and minimal changes to existing ad-exchange architecture. The scalability and privacy issue are address by avoiding server-side data aggregation, which in turn also reduces server's storage and computational requirements. The approach can be easily integrated with existing ad-exchange platforms, and bid optimization frameworks, allowing advertisers highly-granular ad personalization without significant infrastructure investments. **The methodology is currently under scrutiny for patent application**.

Audience Pooling Framework

May 2016 – Oct 2016, Media IQ Digital India Ltd., Bengaluru
Keywords: Map-Reduce, Hadoop Streaming, Data Structures, Design Pattern, Scalable Search Platform, Big Data
Technology: Hadoop Streaming, Java
Number of Team Members: 1

Description: The aim of the project was to develop query engine to filter data from daily feed satisfying any given combination of rules. Since the aim was fast retrieval of records satisfying predefined conditions, I implemented data structures, which despite higher insertion and deletion cost, were fast when it came to retrieval. Storing this data structure in memory of individual mappers, and using Hadoop streaming API, the platform is capable of handling complex queries. Also, the current code can be easily scaled to define new rules owing to the modular code which is built upon design patterns like *Factory Pattern*, *Command Pattern*, and *Chain of Responsibility*. For the ease of usage, I also added a UI on top of the platform for configuring rules. The platform now powers insights which previously were quite difficult to obtain.

Map Reduce Framework for Fast Geo-Spatial Queries (Locate)

Mar 2016 – May 2016, Media IQ Digital India Ltd., Bengaluru

Keywords: Map-Reduce, Data Structures, Hashing, Hadoop Streaming, Big Data

Technology: Hadoop Streaming, Java

Number of Team Members: 1

Description: The location-based user behavior can be captured by pooling in the users who had visited locations in the areas of interest. However, given the sheer volume of data to be processed (around **3.5 TB** daily), and number of locations to be matched (around **45k**), joining the location dataset and input feed proved to be expensive in terms of storage and computation. To address the issue, and put learnings to practice, as a pet project over the weekend, I independently wrote a custom Map-Reduce job for geo-spatial queries, where I converted latitude-longitude-radius triplets to list of Geohashes. This dimensionality reduction (interleaving Latitude, Longitude and Radius) by appropriate hashing technique (Geohash), gave **60x speedup** over hive queries. The custom job reduced the processing time taken **from couple of hours to merely 2 minutes** for the same number of nodes. The job is linearly scalable with respect to size of input feed, and now, is an integral part of all the location based insights and reporting tools at the company.

Deep Learning for Parking Slot Occupancy

Apr 2016 – May 2016, Startup in Stealth Mode, Bengaluru

Keywords: Deep Learning, Computer Vision, Transfer Learning

Technology: TensorFlow, Keras

Number of Team Members: 1

Description: The aim of the project was to develop vision based model for identification of empty parking slot. I used PKLot dataset (Almeida, P., Oliveira, L. S., Silva Jr, E., Britto Jr, A., Koerich, A., PKLot - A robust dataset for parking lot classification, Expert Systems with Applications, 42(11):4937-4949, 2015) for training the model. As preprocessing I resized all the images to 32x32 before proceeding with the training. The model was designed as multinomial classification, and was built upon Google's Inception Model. By using Transfer Learning Technique, I achieved classification accuracy of **96%** on balanced dataset. Further, randomly sampling images with replacement for generation of training batches gave better results as compared to predefined batches, and boosted the accuracy to **98%**.

Pupil Detection for Gaze Tracking

Nov 2015 – Dec 2015, Hackathon Project, Oracle India Pvt. Ltd., Bengaluru

Keywords: Pupil Detection, Support Vector Machines, Kalman Filter

Technology: Python, OpenCV, scikit-learn, pandas

Number of Team Members: 3

Description: The aim of the project was to generate heatmap by tracking user's gaze on screen. We implemented gaze tracking system by implementing pupil capturing using eye center localization as proposed by Timm F. and Erhardt B. in their paper *Accurate Eye Centre Localisation by Means of Gradients VISAPP 11 (2011)*, and later trained SVM model to track user's gaze on screen. The output by SVM model was wavy and to get a smoother distribution, we later added unimodal probability distribution obtained by Kalman Filter. The module developed during the hackathon was later taken for integration by UX Researchers at Oracle to perform visual analysis of attention span for web content

Geo-cleansing of Location Data

Aug 2015 – Feb 2016, Oracle India Pvt. Ltd., Bengaluru

Keywords: Phonetic Algorithm, Soundex, Fuzzy Login

Technology: PLSQL

Number of Team Members: 1

Description: The project aimed at cleansing massive geographical datasets to get rid of typographical errors and duplicate records. I initially found similar records by using *Levenshtein Distance* as similarity matrix, however this led to high number of false positives (about 30% of predicted duplicates). To tighten the false positives, I later implemented fuzzy logic using *Levenshtein Distance*, *Soundex*, and *New York State Identification and Intelligence System*. The accuracy of the model was further enhanced by imposing hierarchy constraints on locations (e.g. only cities in same states would be considered for matching, and the states should have been already cleansed before proceeding with cities). The precision for the tool was **88%**, with miss of **4%**.

Extraction of Bio-Medical Implant

Jan 2015 – May 2015, GlobalLogic India Pvt. Ltd.
Keywords: Biomedical Imaging, Image Processing, Pattern Matching, Morphological Transformations
Technology: MATLAB, OpenCV, Android SDK
Number of Team Members: 1
Description: At GlobalLogic, I worked for one of the top-3 medical technological conglomerate in the world, and worked on development of image processing pipeline for extraction of bio-medical implant from x-ray. The task was preceded by denoising image using median filter, and improving contrast in image by applying adaptive histogram equalization. The denoised-contrast-enhanced image was then passed to morphological filters to find possible candidates for implant, and finally the candidates where narrowed down using pattern matching (on image descriptors). The final decision still to select the position of implant still lied with the operator. The prototyping of the approach was done using MATLAB, and the code was later ported to Android.

Virtual Immersive Cognitive Exercises

May 2014 – Jul 2014, University of Manitoba, Winnipeg
Keywords: VR, Game Engine, Collision Detection, Electro-oculography
Technology: C, C++, Oculus Rift
Number of Team Members: 4 (Guided by Prof. Zahra Moussavi and Dr. Ahmad Byagowi)
Description: Worked on a head mounted tracking system for spatial capability assessment of humans. As part of the project, we developed Virtual Immersive Cognitive Exercises for Alzheimer and Dementia's patients using Oculus Rift as head mounted gear, and VR chair integrated with joystick to move around the virtual world. For stabilizing the rendered content, I also worked on electro-oculography to understand eye-movements while walking with Oculus Rift. We also worked on creation of a few neuro-rehabilitation games for treatment of people who have visual processing problems.

Recreation of Virtual 3D-World from SLAM Map

Jul 2014 – Aug 2014, University of Manitoba, Winnipeg
Keywords: VR, Image Processing, Morphological Transformation
Technology: Python, OpenCV,
Number of Team Members: 2 (Guided by Dr. Ahmad Byagowi)
Description: The project aimed at recreation of virtual 3d-world from the SLAM Map obtained using Laser-SLAM. The task was accomplished by denoising the image by median filter to remove speckles, and Gaussian Blur followed by contour detection. The detected contours were then scaled and used to obtain position of walls to be recreated in Virtual World.

Electronics Aid for Elder and Sick

Jan 2013 – Apr 2013, Study Project, BITS Pilani
Keywords: Embedded System, Assistive Aid, Microcontroller Programming, Wireless Switches, IR Reciever
Technology: C, Arduino, Sensor Interfacing
Number of Team Members: 1 (Guided by Dr. Hari Om Bansal, Associate Professor, BITS Pilani)
Description: The project started as development of S*mart Home* module, and later got focused on development of wireless switches as an aid for elderly and sick people. The choice of infra-red for wireless communication was to avoid interference with other medical devices in hospitals. The task was accomplished by using decoding IR Signal from TV-remote controller, and triggering relays to operate electrical appliances. Since the frequency at which the IR Signals are emitted from remote controller is high, the input from IR receiver were read using lower level of abstractions. The control signals from the remote were easy to identify as the remote controller had distinctive start and end sequence. The work done was published in *IEEE Conference Proceeding of Recent Advances and Innovations in Engineering (ICRAIE), 2014*. The project was further enhanced by integrating it with a web layer, and establishing connection between Arduino and server using serial communication.